

Christophe Jacob · Hassane Zouhal · Jacques Prioux
Arlette Gratas-Delamarche · Danièle Bentué-Ferrer
Paul Delamarche

Effect of the intensity of training on catecholamine responses to supramaximal exercise in endurance-trained men

Accepted: 21 November 2002 / Published online: 12 November 2003
© Springer-Verlag 2003

Abstract In this study we investigated whether plasma catecholamine responses to the Wingate test are affected by the intensity of training in endurance-trained subjects. To do this we compared plasma adrenaline (A) and noradrenaline (NA) concentrations in response to a Wingate test in three different groups: specialist middle-distance runners (MDR) in 800-m and 1,500-m races, specialist long-distance runners (LDR) 5,000-m and 10,000-m races, and untrained subjects (UT). The maximal power ($W_{\max}$) and the mean power (W) were determined from the Wingate test. Blood lactate (La), plasma A and NA concentrations were analysed at rest (La_0 , A_0 and NA_0), immediately at the end of the exercise ($A_{\max}$ and $NA_{\max}$) and after 5 min recovery ($La_{\max}$, A_5 and NA_5). The ratio $A_{\max}/NA_{\max}$ was considered as an index of the adrenal medulla responsiveness to the sympathetic nervous activity. At the end of the test, $W_{\max}$ and W were similar in the three groups but $La_{\max}$ was significantly greater in MDR compared to LDR and UT [15.2 (2.2) mmol l^{-1} , 11.7 (3.1) mmol l^{-1} , 11.6 (1.6) mmol l^{-1} , respectively, for MDR, LDR and UT; mean (SD)]. Concerning the plasma catecholamine concentrations in response to exercise, MDR and LDR $A_{\max}$ values [3.73 (1.53) nmol l^{-1} , 3.47 (0.74) nmol l^{-1} , respectively, for MDR and LDR] were significantly

greater than those of UT [1.48 (0.32) nmol l^{-1}] who also exhibited the lowest $NA_{\max}$ values [11.09 (6.58) nmol l^{-1}] compared to MDR and LDR [20.43 (3.51) nmol l^{-1} ; 15.85 (4.88) nmol l^{-1} , respectively, for MDR and LDR]. However, no significant differences were observed between the two trained groups either for $A_{\max}$ or $NA_{\max}$. These results suggest that long-term endurance training can enhance plasma catecholamine concentrations in response to supramaximal exercise. However, as there were no significant differences between MDR and LDR $A_{\max}$ and $NA_{\max}$ values, the effect of the intensity of training remains to be clarified.

Keywords Adrenaline · Athletes · Noradrenaline · Supramaximal exercise

Introduction

Several studies have reported greater plasma adrenaline concentrations in endurance-trained subjects compared to untrained ones in response to different stimuli (Kjaer et al. 1985, 1986; Kjaer and Galbo 1988) and deal with “sports adrenal medulla” (Kjaer 1998) which correspond to a higher capacity to secrete adrenaline (A) in trained subjects. Nevertheless concerning the type of training sessions which preferentially increase adrenergic responses to supramaximal exercise, data remained conflicting. In fact, while Kjaer et al. (1986) reported greater A responses in endurance-trained subjects compared to untrained ones in response to 2 min at 110% of $\dot{V}O_{2\max}$, Zouhal et al. (2001) did not find any significant differences between endurance-trained and untrained subjects in response to a Wingate test, but reported significant higher plasma A concentrations in sprint-trained subjects. Thus, these authors suggested that endurance-trained subjects tested in the study by Kjaer et al. (1986) may have been more intensively trained. Also in line with this, Strobel et al. (1999) when comparing anaerobic-trained subjects and aerobic-trained

C. Jacob (✉) · H. Zouhal · A. Gratas-Delamarche
P. Delamarche
Laboratoire de Physiologie et de Biomécanique de l'Exercice
Musculaire, UFR-APS, Université de Rennes II,
Av. Charles Tillon, CS 24414, 35044 Rennes Cedex, France
E-mail: christophe.jacob@uhb.fr
Tel.: +33-2-99141775
Fax: +33-2-99141774

J. Prioux
Laboratoire de physiologie et de psychologie de l'exercice,
UFR-STAPS, Université de Nantes, 25 bis bd Guy,
44300 Mollet, France
D. Bentué-Ferrer
Laboratoire de Pharmacologie, Faculté des sciences, Av. Léon
Bernard, 35044 Rennes Cedex, France

subjects also reported greater A concentration in the anaerobic-trained group compared to the aerobic-trained group in response to a 2-min prolonged sprint. These authors also suggested that the intensity of the training sessions rather than training volume is the most important factor able to induce a higher catecholamine response to exercise. In other words, sprint training sessions, which are the more intense training sessions, are the most important factor able to increase A responses to sprint exercise. Therefore, it was of interest to verify this hypothesis. To do this, we compared catecholamine responses to a supramaximal exercise (Wingate test) in two groups of endurance-trained athletes with different competition specialties—middle-distance runners (MDRs; 800 m and 1,500 m) and long-distance runners (LDRs; 5,000 m and 10,000 m), and an untrained group (UT). The two trained groups (MDR and LDR) had the same training volume per week but the MDR training program was more intense than for LDRs. In fact, MDRs include sprint training sessions in their training program, in contrast to LDRs whose training program consists only of aerobic training sessions. It can then be assumed that the MDR training program differs significantly from the LDR one in regard to intensity.

In this study, the Wingate test was chosen as it has been reported that the catecholamine responses reach their highest values immediately after this type of exercise more than in response to aerobic exercise (Kindermann et al. 1982; Schwarz and Kinderman 1990) or to other stimuli such as glucagon ingestion (Kjaer and Galbo 1988) or mental stress (Forsman and Lindblad 1983). The maximal plasma A concentration ($A_{\max}$) determined immediately at the end of supramaximal exercise can also be considered as an index of the maximal adrenal medulla responsiveness (Kjaer 1998; Zouhal et al. 1998; Jacob et al. 2002) and the maximal plasma noradrenaline concentration ($NA_{\max}$) as an index of the maximal sympathetic nervous activity (Yamaguchi and Kopin 1979). To investigate the responsiveness of the adrenal medulla to the sympathetic nervous input, we compared the A/NA ratio between the two groups (Zouhal et al. 1998, 2001; Jacob et al. 2002).

Methods

Subjects

Nineteen subjects gave their informed written consent to take part in this study. There were two groups of endurance-trained subjects: one composed of 6 MDRs, specialists in 800-m and 1,500-m races and the other of 6 LDRs, specialists in 5,000-m and 10,000-m races. The trained subjects were all competing at high national level and had each trained for 10–12 h per week for at least 4 years. Moreover, it is important to note that all the training programs for these subjects were planned for at least 3 years by two members of our laboratory who coach runners, so that we could attest to the training content and specificity. Seven untrained subjects (UT) were recruited as they had not undertaken more than 2 h of physical activity per week for at least 3 years.

Training sessions

In this study all the subjects were endurance trained but as the anaerobic contribution in middle-distance running is greater than in long-distance running, the MDR training program consisted of aerobic and anaerobic training sessions. MDRs trained for maximal speed running (anaerobic training; once a week), prolonged sprint (anaerobic training; twice a week) and undertook aerobic training three times a week in order to increase their maximal oxygen consumption ($\dot{V}O_{2\max}$). In contrast, LDRs undertook only aerobic training sessions three times a week in order to increase their $\dot{V}O_{2\max}$ () and maximal aerobic capacity.

Procedure

On day 1 the percentage of body fat (Fat) was estimated from four skinfold thicknesses according to the method of Durnin and Rahaman (1976). The lean body mass (LBM) was calculated by subtracting the total body fat from the body weight. $\dot{V}O_{2\max}$ was measured during an incremental exercise test performed on a treadmill using a breath-by-breath automated exercise metabolic system (CPX; Medical Graphics, St. Paul, Minn., USA).

On day 2 the force-velocity test (F/v test) was performed using the technique of Vandewalle et al. (1985). This test, using an Ergomeca bicycle, consisted of a succession of several supramaximal bouts of roughly 6 s against loads increased by 1 kg after each bout until inability to pursue the test. A complete recovery (5 min) was allowed between successive bouts. The velocity was recorded using a photoelectric cell fixed on the wheel of the cycle and connected to a computer. Only the highest speed value was recorded for each load. By multiplying load and speed values, a power curve was then built at each bout until the maximum. The load ($F_{\max}$) corresponding to this maximal power was then used for a Wingate test performed on the same day after a 30-min recovery in order to familiarise the subjects with this experiment.

The last part of the study took place on day 3 in the absence of any competition or exhaustive physical activity. The experiment started for all subjects at about 9.00 a.m., 2 h after a standardised breakfast. On arrival each subject lay down on a bed and a heparinised catheter was inserted into an antecubital vein. The subject mounted the bicycle after a 30-min rest, and the first blood sample (5 ml) was drawn to determine resting plasma A (A_0) and NA (NA_0) levels. Simultaneously, a few microlitres of arterialed capillary blood were drawn to determine resting blood lactate (La_0). A warm-up was allowed for 15 min at a submaximal power of about 50–60 W at a very low velocity. Subsequently, the Wingate test was performed. During this exercise the subject was asked to cycle for 30 s as fast as possible on the same machine and against the previously determined load ($F_{\max}$). The velocity was again recorded throughout the test. The power produced could be calculated at any time with the highest value chosen for maximal power ($W_{\max}$). The average of all measured power values during the 30-s bout was taken as mean power W .

Two more venous blood samples were again drawn (5 ml), one immediately after the cycling was stopped to analyse $A_{\max}$ and $NA_{\max}$. As the half-life of catecholamine is very short, this blood sample was taken as fast as possible, always by the same technician and using the same standardised procedure. A final blood sample was taken after 5 min recovery (A_5 and NA_5). To determine the maximal blood lactate ($La_{\max}$), arterialed capillary blood was collected 3 min after the end of the exercise.

Blood analysis

Lactate was immediately determined enzymatically in the capillary blood sample using a lactate analyser (Microzym, Cetrix). Plasma from the venous blood samples was separated by centrifugation at 3,000 g and stored at -80°C for subsequent catecholamine analysis. Plasma A and NA concentrations were evaluated by high-performance liquid chromatography, following the method of Koubi et al.

(1991): before the HPLC run the catecholamines were extracted by selective adsorption to aluminium oxide (Chromsystems-HPLC-Kit; Waters, Milford, Mass., USA). The aluminium oxide was shaken briefly in extraction buffer (50 µl) and then 1 ml of plasma was added with 50 µl internal standard solution (600 pg dihydroxybenzylamine). The aluminium oxide was then washed three times, with a brief centrifugation between washes. The catecholamines were extracted with 120 µl elution buffer by shaking briefly and subjecting it to a final centrifugation at 1,500 g for 1 min, then 50 µl of the sample eluant was injected into HPLC column (Resolve TM 5 µl Sherical C 18, HPLC column, Waters) and eluted with a mobile phase. The flow rate was 1 ml min⁻¹ at 13.8 mPa and a potential of 0.60 V. The chromatogram was analysed by computer integration (Baseline 815; Waters).

Calculations

To test the responsiveness of the adrenal medulla to the sympathetic nervous activity the ratio A/NA was calculated at rest (A_0/NA_0) and immediately at the end of the Wingate test ($A_{\max}/NA_{\max}$) (Zouhal et al. 1998). The disappearance of A and NA was determined by the ratio $(A_{\max}-A_5)/A_{\max}$ and $(NA_{\max}-NA_5)/NA_{\max}$ (Jacob et al. 2002).

Statistical tests

Data are summarised as mean (SD). Statistical evaluation of differences between UT, MDR and LDR was made by an ANOVA on Ranks. In the event of significant differences, the significance was evaluated with a Mann-Witney's test. A value of $P < 0.05$ was accepted as the minimal level of statistical significance.

Results

Morphological and physiological characteristics

As shown in Table 1 the three groups were matched for age, weight, height and LBM. $\dot{V}O_{2\max}$ related to LBM were statistically greater in trained subjects compared to UT and in LDR compared to MDR.

Performances and maximal blood lactate during the Wingate test

Table 2 shows that $W_{\max}$ and W expressed in absolute or relative values were about the same in the three groups. $La_{\max}$ values were significantly greater in MDR compared to LDR and UT.

Catecholamine responses to the Wingate test

As shown in Table 3 there were no significant differences between MDR and LDR concerning catecholamine concentrations at rest and at the end of the Wingate test, but $A_{\max}$ values were significantly lower in UT compared to MDR and LDR. UT also exhibited the lowest $NA_{\max}$ values but the ratio $A_{\max}/NA_{\max}$ was very similar in the three groups.

After 5 min recovery, NA_5 was significantly higher in MDR compared to LDR but the elimination index of these hormones was not significantly different [A elimination index: 0.7 (0.09); 0.7 (0.02); 0.6 (0.23), respectively, for MDR, LDR and UT; NA elimination index: 0.4 (0.22); 0.5 (0.12); 0.4 (0.26), respectively, for MDR, LDR and UT).

Discussion

In the present study and that previously reported by Kjaer (1998), maximal plasma A concentrations in response to the Wingate test were significantly greater in endurance-trained subjects (MDR and LDR) compared to untrained ones (UT). However, in contrast to our attempts, we have not found any significant differences in maximal plasma A and NA concentrations in response to the Wingate test between the two endurance-trained groups MDR and LDR, despite significant differences in their training program intensity, $\dot{V}O_{2\max}$ and $La_{\max}$.

Morphological and physiological characteristics

In our study the three groups were about the same age, weight and height. The fat-free mass of the three groups was very close, even though UT had a significantly higher percentage fat. As expected $\dot{V}O_{2\max}$ for MDR and LDR was statistically higher than for UT, and LDR exhibited higher values than MDR. These differences in $\dot{V}O_{2\max}$ can easily be explained: LDR trained more in order to increase $\dot{V}O_{2\max}$ than MDR since the contribution of aerobic pathways is more important during long-distance (5,000 m and 10,000 m) races compared to middle distance (800 m and 1,500 m). In fact, during races of 800–1,500 m, the aerobic pathway is known to

Table 1 Morphological characteristics and aerobic capacity of the three groups. Values are mean (SD). *LBM* Lean body mass, *LDR* long-distance runner, *MDR* middle-distance runner, *UT* untrained, $\dot{V}O_{2\max}$ maximum O₂ consumption

	Age (years)	Height (cm)	Weight (kg)	Fat (%)	LBM (kg)	$\dot{V}O_{2\max}$ (ml min ⁻¹ kg ⁻¹)	$\dot{V}O_{2\max}$ (ml min ⁻¹ kg LBM ⁻¹)
MDR (<i>n</i> = 6)	21.2 (1.5)	173 (6.9)	64.8 (5.8)	9.1 (1.5)	58.9 (6.1)	65.0 (3.2)	72.8 (5.1)*
LDR (<i>n</i> = 6)	21.1 (2.3)	178 (6.5)	65.0 (6.4)	11.7 (2.3)	58.0 (6.6)	70.5 (4.2)	79.2 (4.2)
UT (<i>n</i> = 7)	21.3 (1.3)	175 (6.9)	67.9 (6.9)	15.5 (3.7)** ***	50.6 (7.1)	43.7 (7.1)** ****	51.5 (7.2)** ****

*Significant difference between MDR and LDR, $P < 0.05$

Significant difference between MDR and UT, $P < 0.05$, * $P < 0.01$

Significant difference between UT and LDR, $P < 0.05$, * $P < 0.01$

Table 2 Performances determined during the Wingate test and maximal blood lactate. Values are mean (SD). La_{max} Maximal blood lactate, W mean power, W_{max} maximal power

	W_{max}			W			La_{max}
	W	W kg ⁻¹	W kg LBM ⁻¹	W	W kg ⁻¹	W kg LBM ⁻¹	mmol l ⁻¹
MDR	921 (117)	14.5 (0.5)	15.8 (0.5)	700 (98)	11.0 (1.9)	12.2 (1.0)	15.2 (2.2)*
LDR	833 (85)	12.8 (0.8)	14.4 (1.0)	625 (77)	9.6 (0.9)	10.8 (1.1)	11.7 (3.1)
UT	896 (134)	13.1 (2.4)	15.6 (2.4)	675 (79)	10.1 (1.8)	11.5 (1.1)	11.6 (1.6)***

*Significant difference between MDR and LDR, $P < 0.05$ **Significant difference between MDR and UT, $P < 0.05$, *** $P < 0.01$ **Table 3** Plasma adrenaline (A) and nordrenaline (NA) concentrations determined at rest (A_0 and NA_0), immediately after the Wingate test (A_{max} and NA_{max}) and after 5 min recovery (A_5 and NA_5)

	A_0 (nmol l ⁻¹)	A_{max} (nmol l ⁻¹)	A_5 (nmol l ⁻¹)	NA_0 (nmol l ⁻¹)	NA_{max} (nmol l ⁻¹)	NA_5 (nmol l ⁻¹)	A_{max}/NA_{max}
MDR	0.54 (0.08)	3.73 (1.53)	0.93 (0.43)	2.75 (0.81)	20.43 (3.51)	11.03 (4.33)	0.186 (0.07)
LDR	0.79 (0.36)	3.47 (0.74)	0.83 (0.29)	1.71 (0.64)	15.85 (4.88)	4.97 (1.36)	0.229 (0.09)
UT	0.38 (0.18)****	1.48 (0.32)***	0.93 (0.60)	2.37 (0.55)	11.09 (6.58)**	6.62 (3.2)	0.171 (0.07)

*Significant difference between MDR and LDR, $P < 0.05$ **Significant difference between UT and MDR, $P < 0.05$, *** $P < 0.01$ ****Significant difference between UT and LDR, $P < 0.05$, ***** $P < 0.01$

contribute about 50–60% of the energetic expenditure, whereas this could be at least 95% for a 5,000-m race and even more for 10,000 m (Lacour et al. 1990). Moreover it is important to note that the $\dot{V}O_{2max}$ -related values were very high in the two trained groups. These results are in accordance with those of previous studies (Costill et al. 1976; Legaz and Ostari 2000) and confirm the high endurance training level of the two endurance-trained groups.

Performances and La_{max} during the Wingate test

Mean values of the performances determined during the Wingate test were higher in MDR but not significantly different from those of LDR and UT, even though MDR performed some sprint training sessions. Since MDR did not perform sprint training more than three times a week and only during part of the year, the effect of these training sessions on cycling performances is likely to be limited. Nevill et al. (1989) reported an increase in W_{max} and W determined during a Wingate test after sprint training. However, in this study all the subjects were training more often (three to four times a week) and especially in order to increase their performances in the Wingate test (brief and prolonged sprint cycling exercise). Nevertheless, MDR La_{max} concentrations were significantly greater than those of LDR and UT, which is in accordance with their higher volume of anaerobic training. This result argues in favour of a higher glycolytic capacity and/or stimulation during the test. In fact, sprint training sessions can increase glycolytic capacity by enhancing glycolytic enzyme activity and/or concentrations. But we can also hypothesise that this higher La_{max} is due to a higher stimulation of

glycogenolysis by the catecholamines. In fact, MDR exhibited the highest mean A_{max} when A is known to directly stimulate glycogenolysis (Richter et al. 1981; Stainsby et al. 1985) and A concentrations correlate with lactate values (Weltmann et al. 1994). Collomp et al. (1991) reported higher lactate in response to higher catecholamine concentrations induced by a caffeine ingestion without any significant increase in performances, and then suggested that other factors like H^+ and P_i accumulation may cancel the positive effect of the higher glycogenolysis stimulation. Also in line with this, Nevill et al. (1989) reported that in their study, the subjects with the highest plasma A concentrations after the 30-s sprint also had the highest buffering capacity.

Our study supports that of Stainsby et al. (1985) suggesting that catecholamine stimulates glycogenolysis in tissues other than working muscles, since MDR performances were not significantly different from those of LDR and UT.

Finally, the results concerning MDR and LDR performances, La_{max} and $\dot{V}O_{2max}$ confirm the differences between their training program intensity and are relevant to previous work (Taunton et al. 1981).

Catecholamine responses to the Wingate test:

In this study the endurance-trained subjects (MDR and LDR) exhibited significant higher A_{max} values compared to UT, who also exhibited the lowest NA_{max} values. Those results, greater plasma A concentrations in response to exercise in endurance-trained subjects, are in accordance with previous studies using other stimuli (Kjaer et al. 1986, 1992; Kjaer and Galbo 1988; Silvermann and Mazzeo 1996) and argued in favour of

so-called "sports adrenal medulla" which corresponds to a higher capacity to secrete A in response to exercise (Kjaer 1998). In fact, it has been suggested that this higher capacity to secrete A can be reflected by significantly higher A concentrations in endurance-trained athletes (Kjaer et al. 1986; Kjaer and Galbo 1988). This may be explained by an increase of the adrenal medulla volume and an increase of the A content of these glands as reported in rats (Stallknecht et al. 1990; Schmidt et al. 1992). In our study, untrained subjects also exhibited the lowest NA_{max} values. Higher plasma NA concentrations in response to exercise in trained subjects have already been reported after endurance training by Greiwe et al. (1999) or when comparing endurance-trained subjects to untrained ones (Silvermann and Mazzeo 1996). In their study Greiwe et al. (1999) incriminated the absolute work load which was increased after training. This hypothesis cannot be included in our study since performances determined during the Wingate test were not significantly different between the three groups. However, like Greiwe et al. (1999) when talking about the effect of training or Lehmann and Keul (1986) when referring to the effect of aging, we can hypothesise that these higher NA_{max} values may be caused by a compensatory mechanism which may depend on the decreased sensitivity of the organism to beta-adrenergic stimulation possibly induced by training.

Moreover, in our study and in contrast to our attempts no significant differences were observed between MDR and LDR concerning A_{max} and NA_{max} . Basal values (A_0 and NA_0) and elimination index ($(A_{max}-A_5)/A_{max}$ and $(NA_{max}-NA_5)/NA_{max}$) were also not significantly different between these two groups. Then we could hypothesise that A and NA secretion may have been about the same in the two groups with different competing specificity. The two groups were competing at the same high level, and we could attest that their training intensities were different (their coaches are laboratory members). Two main hypotheses can then be considered to explain the lack of significant differences concerning plasma catecholamine concentrations in response to supramaximal exercise between these two groups:

1. Anaerobic training is not a major factor able to induce an increase of plasma A in response to exercise. This hypothesis is not probably true. In fact, previous studies reported significantly higher A responses in sprint-trained subjects compared to untrained ones in response to a Wingate test (Zouhal et al. 1998) or significantly higher A responses in anaerobic-trained subjects compared to aerobic-trained ones in response to a 2-min prolonged sprint (Strobel et al. 1999).
2. Anaerobic training is able to increase A responses to sprint exercise but MDR sprint training volume was not enough to induce such adaptations, which may be hard to attain. This hypothesis can be considered since in our study and even if MDR and LDR

training programs were different concerning the sprint training session volume, the W_{max} and W of the two groups were relatively close despite sprint training being known to enhance sprint performances. Moreover, it is important to note that the data concerning performances were markedly lower than those reported previously in sprint-trained subjects (Zouhal et al. 1998). Concerning the catecholamine responses we can also note that the MDR and LDR plasma A concentrations were twofold lower than those of the sprinter values that we measured in response to the same test and with the same protocol (Zouhal et al. 1998). The most important differences between sprint-trained subjects and MDR are the number of sprint-training sessions and the number of strength-training sessions. Guezennec et al. (1986) suggested that strength training has no effects on catecholamine responses to exercise. Therefore it is reasonable to assume that sprint-training sessions are able to increase plasma A responses to sprint exercise. Then the lack of significant differences between MDR and LDR concerning A_{max} can be explained by the relative low proportion of sprint-training sessions in the MDR training program. In fact, MDRs do not perform sprint-training sessions all year like sprinters. However, as the A responses of the two groups were very close and as we could attest differences between MDRs and LDRs concerning sprint-training session volume, we can hypothesise that the twofold higher A responses in sprinters we reported in a previous study in response to the same test are more likely due to genetic predisposition as suggested in endurance-trained subjects by Kjaer et al. (1992).

In conclusion, our results indicated that in response to a Wingate test and as reported before in response to various submaximal or maximal exercises, and or other stimuli, endurance-trained subjects exhibited higher catecholamine responses to exercise than untrained ones. This study also suggested that the volume of sprint-training sessions may be an important factor for enhancing the catecholamine responses to exercise.

Acknowledgements The authors wish to thank Dominique Paul and Marie-Thérèse Gougeon for technical assistance. The Ethical Committee of Rennes 1 approved the entire procedure (Comité Consultatif de Protection des Personnes en Recherche Biomédicale: CCPPRB).

References

- Collomp K, Ahmaidi S, Audran M, Chanal J-L, Préfaut C (1991) Effects of caffeine ingestion on performance and anaerobic metabolism during the Wingate test. *Int J Sports Med* 12:439-443
- Costill DL, Fink WJ, Pollock ML (1976) Muscle fiber composition and enzyme activities of elite distance runners. *Med Sci Sports Exerc* 8:96-100
- Durnin J, Rahaman M (1976) The assessment of the amount of fat in the human body from measurements of skinfold thickness. *Br J Nutr* 21:681-689

- Forsman L, Lindblad LE (1983) Effect of mental stress on baroreceptor-mediated changes in blood pressure and heart rate and on plasma catecholamines and subjective responses in healthy men and women. *Psychosom Med* 45:435–445
- Greiwe JS, Hickner RC, Shah SD, Cryer PE, Holloszy JO (1999) Norepinephrine response to exercise at the same relative intensity before and after endurance exercise training. *J Appl Physiol* 86:531–535
- Guezennec CY, Leger L, Lhoste F, Aymonod M, Pesquies PC (1986) Hormone and metabolite response to weight-lifting training sessions. *Int J Sports Med* 7:100–105
- Jacob C, Zouhal H, Vincent S, Gratas-Delamarche A, Berthon PM, Bentue-Ferrer D, Delamarche P (2002) Training status (endurance or sprint) and catecholamine response to the Wingate test in women. *Int J Sports Med* 23:342–347
- Kindermann W, Schnabel A, Schmitt WN, Biro G, Cassens J, Weber F (1982) Catecholamine, growth hormone, cortisol, insulin, and sex hormones in anaerobic and aerobic exercise. *Eur J Appl Physiol* 49:389–399
- Kjaer M (1998) Adrenal medulla and exercise training. *Eur J Appl Physiol* 77:195–199
- Kjaer M, Galbo H (1988) The effect of physical training on the capacity to secrete epinephrine. *J Appl Physiol* 64:11–16
- Kjaer M, Christensen NJ, Sonne B, Richter EA, Galbo H (1985) Effect of exercise on epinephrine turnover in trained and untrained male subjects. *J Appl Physiol* 59:1061–1067
- Kjaer M, Farrell PA, Christensen NJ, Galbo H (1986) Increased epinephrine response and inaccurate glycoregulation in exercising athletes. *J Appl Physiol* 61:1693–1700
- Kjaer M, Mikines KJ, Linstow MV, Nicolaisen T, Galbo H (1992) Effect of 5 weeks of detraining on epinephrine response to insulin-induced hypoglycemia in athletes. *J Appl Physiol* 72:1201–1204
- Koubi HE, Desplanché D, Gabrielle C, Cottet-Emard JM, Semporer B, Favier R (1991) Exercise endurance and fuel utilisation: a reevaluation of the effects of fasting. *J Appl Physiol* 70:1337–1343
- Lacour JR, Padilla-Magunacelaya S, Barthelemy JC, Dormois D (1990) The energetics of middle-distance running. *Eur J Appl Physiol* 60:38–43
- Legaz AA, Ostariz ES (2000) Maximal oxygen consumption among homogeneous groups of elite athletes in running events. *Sci Sport* 15:318
- Lehmann M, Keul J (1986) Age-associated changes of exercise-induced plasma catecholamine responses. *Eur J Appl Physiol* 55:302–306
- Nevill ME, Boobis LH, Brooks S, Williams C (1989) Effect of training on muscle metabolism during treadmill sprinting. *J Appl Physiol* 67:2376–2382
- Richter EA, Sonne B, Christensen NJ, Galbo H (1981) Role of epinephrine for muscular glycogenolysis and pancreatic hormonal secretion in running rats. *Am J Physiol* 240:E526–E532
- Schmidt KN, Gosselin LE, Stanley WC (1992) Endurance exercise training causes adrenal medullary hypertrophy in young and old Fisher 344 rats. *Horm Metab Res* 24:511–515
- Schwarz L, Kindermann W (1990) β -Endorphin, adrenocorticotrophic hormone, cortisol and catecholamines during aerobic and anaerobic exercise. *Eur J Appl Physiol* 61:165–171
- Silverman HG, Mazzeo RS (1996) Hormonal responses to maximal and submaximal exercise in trained and untrained men of various ages. *J Gerontol* 51A:B30–B37
- Stainsby WN, Sumners C, Eitzman PD (1985) Effects of catecholamines on lactic acid output during progressive working contractions. *J Appl Physiol* 59:1809–1814
- Stallknecht B, Kjaer M, Ploug T, Maroun L, Ohkuwa T, Vinten J, Mikines KJ, Galbo H (1990) Diminished epinephrine response to hypoglycemia despite enlarged adrenal medulla in trained rats. *Am J Physiol* 259:R998–R1003
- Strobel G, Friedmann B, Siebold R, Bärtsch P (1999) Effect of severe exercise on plasma catecholamines in differently trained athletes. *Med Sci Sports Exerc* 31:560–565
- Taunton JE, Maron H, Wilkinson JG (1981) Anaerobic performance in middle and long distance runners. *Can J Appl Sports Sci* 6:109–113
- Vandewalle H, Perès G, Heller J, Monod H (1985) All-out anaerobic capacity tests on cycle ergometers. *Eur J Appl Physiol* 54:222–229
- Weltman A, Wood CM, Womach CJ, Davis SE, Blumer JL, Alvarez J, Sauer K, Gaesser GA (1994) Catecholamine and blood lactate responses to incremental rowing and running exercise. *J Appl Physiol* 76:1144–1149
- Yamaguchi I, Kopin IJ (1979) Plasma catecholamine and blood pressure responses to sympathetic stimulation in rats. *Am J Physiol* 237:H305–H310
- Zouhal H, Rannou F, Gratas-Delamarche A, Monnier M, Bentué-Ferrer D, Delamarche P (1998) Adrenal medulla responsiveness to the sympathetic nervous activity in sprinters and untrained subjects during a supramaximal exercise. *Int J Sports Med* 19:172–176
- Zouhal H, Jacob C, Gratas-Delamarche A, Rannou F, Bentué-Ferrer D, Delamarche P (2001) Effect of training status on the sympatho-adrenal activity during a supramaximal exercise in human. *J Sports Med Phys Fitness* 41:330–336